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Bone punches

Abstract

Disclosed is a disassemblable bone punch comprising a shaft and a handle, wherein the shaft comprises a first shaft part and a second shaft part, wherein the first shaft part comprises at least one first guiding element, and the second shaft part comprises at least one second guiding element, wherein the first shaft part is connected to the second shaft part such that the first guiding element is in engagement with the second guiding element and the first shaft part is movable parallel to a proximal-to-distal direction relative to the second shaft part, wherein the shaft is movable into a disassembly position, into a neutral position and into a punching position.

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Background/Summary

FIELD OF THE INVENTION

(1) The invention relates to a disassemblable bone punches.

STATE OF THE ART

(2) Bone punches are known from the state of the art and are used for the surgical removal of bone, ligaments, cartilage and related tissue in various medical disciplines. As high-priced special products, they are designed for multiple use and must be cleaned after each operation. To support the cleaning process, the bone punches can also be designed to be disassembled. Disassemblable bone punches are also known from the prior art, but compared to non-disassemblable bone punches, they have a large number of additional components which are intended on the one hand to ensure disassembly security, i.e. safe handling during surgical use, and on the other hand to enable them to be disassembled. The disassembly itself, and in particular the release of the disassembly security, then takes place in several steps. In most cases, it is already necessary to remove components for this step. A certain sequence of steps must also be followed when releasing the disassembly security, otherwise the bone punches cannot be disassembled or may even be damaged. In addition, the components used for securing take up extra space, cause additional weight and therefore hinder or complicate the regular, i.e. operational, use of the bone punches.

REPRESENTATION OF THE INVENTION

(3) It is thus the object of the present invention to provide a disassemblable bone punches which has good and safe handling properties and is robust and easy to disassemble.

(4) The object is solved by a disassemblable bone punches according to claim 1. Further features developing the invention are subject of the dependent claims.

(5) A disassemblable bone punch according to the invention comprises a shaft and a handle, wherein the shaft comprises a first shaft part and a second shaft part, wherein the first shaft part comprises at least one first guiding element, and the second shaft part comprises at least one second guiding element, wherein the first shaft part is connected to the second shaft part such that the first guiding element is in engagement with the second guiding element and the first shaft part is movable parallel to a proximal-to-distal direction relative to the second shaft part, wherein the shaft is movable into a disassembly position, into a neutral position and into a punching position, wherein the handle comprises a first handle portion and a second handle portion, wherein the first handle portion is movably connected to the second shaft part, wherein the second handle portion is connected to the second shaft part, wherein by a first movement of the first handle portion the shaft is displaceable from the neutral position into the punching position, characterized in that by a second movement of the first handle portion the shaft can be moved from the neutral position into the disassembly position.

(6) Thus, a disassemblable bone punches is provided, which has a good and safe handling and is robust and easy to disassemble and has the lowest possible number of components.

(7) The neutral position is the position in which the bone punches is in the operating state without any external force being applied.

(8) The punching position is the position to which the bone punches is moved during its main use in order to punch the tissue to be removed. For this purpose, the shaft may have a punching element and a punch intake, whereby during punching, i.e. when the shaft is moved into the punching position, the punching element moves into the punch intake while the tissue to be removed is located between the two elements. The tissue to be removed is then removed or separated by further retraction of the punching element into the punch intake.

(9) The disassembly position is the position in which the shaft and preferably the entire bone punches can be separated into its individual parts. In this position, the first shaft part can be separated from the second shaft part and the first handle portion can be separated from the second handle portion.

(10) The proximal-distal direction is the direction from the proximal end, i.e. the end closest to the user, to the distal end, i.e. the end furthest away from the user, during regular use of the bone punches.

(11) The bone punches is preferably configured in such a way that the first movement moves the

first handle portion towards the second handle portion. Preferably, the bone punches is also configured in such a way that the second movement can be performed away from, in particular opposite to, the first movement.

(12) Preferably, the first handle portion is connected to the second shaft part via a connecting device and the corresponding movability is enabled by the connecting device.

(13) Preferably, the first handle portion is connected to the second shaft part in such a way that the first handle portion can be rotated about a point located in the vicinity or in the area of the second shaft part.

(14) Preferably, the first handle portion is further arranged in such a way that the first handle portion, preferably its head region, touches the first shaft part.

(15) Preferably, the connection between the second handle portion and the second shaft part is fixed in the entire operating state and is also not intended to be detachable for disinfection, i.e. the second handle portion is not movable relative to the second shaft part. The connection can be a non-detachable joint, but also a detachable connection, such as a plug-in or screw connection.

(16) Preferably, the first shaft part comprises a locking area, wherein the locking area is movable from a neutral status to an unlocking status, wherein the locking area is configured such that, when the locking area is in the neutral status, moving of the shaft into the disassembly position is prevented and, by the second movement, the locking area is simultaneously moved into the unlocking status and the shaft is moved into the disassembly position. Thus, it can be prevented that the shaft is unintentionally moved into the disassembly position, which may increase the safety of the bone punches. The locking area can be configured as a locking mechanism.

(17) Preferably, the locking area comprises a deflectable spring area, wherein the locking area is movable into the unlocking status by deflecting the spring area. This simplifies the disassembly of the bone punches.

(18) Preferably, the spring area is configured to be deflectable substantially perpendicular to the proximal distal direction. Thus, it can be prevented that during regular use, while the shaft part is moved parallel to or along the proximal-distal direction, the locking area is unintentionally moved into the unlocking status, which may increase the safety of the bone punches.

(19) Preferably, the spring area is deflectable substantially parallel to a horizontal direction. Thus, a low height of the bone punches can be achieved in a vertical direction, which increases their handling properties. The horizontal direction is located in a horizontal plane and is aligned perpendicular to the proximal-distal direction. The horizontal plane results with respect to the orientation of the bone punches during regular operation or during their main use. The horizontal plane is oriented substantially parallel to the shaft and perpendicular to the handle.

(20) The vertical direction is oriented perpendicular to the proximal-distal direction and perpendicular to the horizontal direction.

(21) Preferably, the spring area is deflectable substantially parallel to a vertical direction. Thus, a small width of the bone punches can be achieved in a horizontal direction, which may increase the handling properties of the same.

(22) Preferably, the spring area comprises a loose end and a fixed end, with the loose end comprising the greatest deflectability and the fixed end being non-deflectable.

(23) Preferably, the spring area comprises a recess, which is in particular rounded or curved and is preferably arranged at the fixed end of the spring area. This prevents stress cracks from forming in the spring area when it is deflected, particularly in the case of regular repeated deflections, which may increase the lifetime of the bone punches. Here, the recess acts as a kind of undercut.

(24) Preferably, the second shaft part comprises a fixing area, the fixing area being configured and arranged to interact with the locking area such that if the locking area is in the neutral status, additional resistance is provided by the fixing area opposing the movement of the shaft into the disassembly position. Thus, it can be additionally prevented that the locking area is unintentionally moved into the unlocking position, whereby the safety of the bone punches can be additionally

increased. In this context, interacting is understood to be preferably the corresponding interaction of the fixing area with the locking area.

(25) Preferably, the locking area comprises at least one first resistance element and/or the fixing area comprises at least one second resistance element, wherein the first resistance element and/or the second resistance element is/are arranged and configured such that an additional resistance opposes the movement of the locking area into the unlocking status. Thus, the safety of the bone punches can be additionally increased.

(26) Preferably, the disassemblable bone punches comprises a first resistance element and a second resistance element, wherein the first resistance element and the second resistance element are interactively arranged and configured such that additional resistance opposes the movement of the locking area into the unlocking status. Thus, the safety of the bone punches can be additionally increased.

(27) Preferably, the first resistance element is configured as a projection and/or the second resistance element is configured as a bulge. Thus, the manufacturing of bone punches can be simplified and their robustness can be increased.

(28) According to a preferred alternative, the first resistance element is configured as a bulge and/or the second resistance element is configured as a cylinder. Thus, the manufacturing of the bone punches can be simplified and their robustness can be increased.

(29) Preferably, the bone punches further comprises a first stop element preferably arranged in the second shaft part, wherein the first stop element is configured and further arranged such that the movability of the first handle portion is limited proximally. Thus, it can be prevented that the locking area is deflected further or loaded more than necessary or intended, which may increase the lifetime and safety of the bone punches.

(30) Preferably, the bone punches further comprises a second stop element preferably arranged in the second shaft portion, wherein the second stop element is configured and further arranged such that the movability of the first handle portion is limited distally. Thus, it can be prevented that the shaft or the first shaft part is deflected further than the punching position indicates or results from this punching position, whereby the lifetime and the safety of the bone punches can be increased.

(31) Preferably, the bone punches further comprise a return member preferably arranged in the second shaft part, wherein the return member is configured and further arranged such that after performing the first movement, the shaft is automatically displaced from the punching position back to the neutral position and, preferably the first movement is opposed by a resistance, preferably an increasing resistance. Thus, during the main use of the bone punches, no force has to be applied by the operator to reset or move the shaft from the punching position to the neutral position, which can increase the handling properties. Furthermore, this can easily provide feedback to the operator regarding the current position or positioning of the shaft during the main use of the bone punches. The increasing resistance may be achieved by the return member having a compression spring. The resistance is increasing in that it increases as the first movement progresses, in particular linearly. In particular, the resistance starts to increase immediately after or in the neutral position. I.e. in the neutral position the resistance is lowest and immediately before the end of the first movement or at the end of the first movement, i.e. in the punching position, the resistance is highest.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1a shows a bone punches with a proximally arranged locking area in a punching position.
- (2) FIG. 1b shows the bone punches displayed in FIG. 1a in a neutral position.
- (3) FIG. 1c shows the bone punches displayed in FIGS. 1a and 1b in a disassembly position.

- (4) FIG. 1d shows the bone punches displayed in FIGS. 1a to 1c in a top view.
- (5) FIG. 1e shows the bone punches displayed in FIGS. 1a to 1d in the neutral position displayed in FIG. 1b in a sectional view.
- (6) FIG. 2a shows a side view of a first shaft part of the bone punches displayed in FIGS. 1a to 1e.
- (7) FIG. 2b shows a bottom view of the first shaft part displayed in FIG. 2a.
- (8) FIG. 2c shows a side view of a second shaft part of the bone punches displayed in FIGS. 1a to 1e.
- (9) FIG. 2d shows a top view of the second shaft part displayed in FIG. 2c.
- (10) FIG. 2e shows a first handle portion of the bone punches displayed in FIGS. 1a to 1e.
- (11) FIG. 2f shows a handle head of the first handle portion displayed in FIG. 2e in a side view.
- (12) FIG. 2g shows a front view of the first handle portion displayed in FIG. 2e.
- (13) FIG. 2h shows a side view of a second handle portion of the bone punches displayed in FIGS. 1a to 1e.
- (14) FIG. 2i shows the second handle portion displayed in FIG. 2h in an oblique top view.
- (15) FIG. 3a shows another bone punches with a distally arranged locking area in a punching position.
- (16) FIG. 3b shows the bone punches displayed in FIG. 3a in a neutral position.
- (17) FIG. 3c shows the bone punches displayed in FIGS. 3a and 3b in a disassembly position.
- (18) FIG. 4a shows a side view of a first shaft part of the bone punches displayed in FIGS. 3a to 3c.
- (19) FIG. 4b shows the first shaft part displayed in FIG. 4a from below.
- (20) FIG. 4c shows a side view of a second shaft part of the bone punches displayed in FIGS. 3a to 3c.
- (21) FIG. 4d shows a top view of the second shaft part displayed in FIG. 4c.
- (22) FIG. 5a shows an exploded view of a connecting device of the bone punches displayed in FIGS. 1a to 1e and 3a to 3b, respectively.
- (23) FIG. 5b shows the connecting device displayed in FIG. 5a in the assembled state.
- (24) FIG. 5c shows a section of a second shaft part with a first stop element in a top view.
- (25) FIG. 5d shows a section of a second shaft part with a second stop element in a top view.
- (26) FIG. 5e shows a return member in a side view.
- (27) FIG. 5f shows a side view of a spring associated with the return member displayed in FIG. 5e.
- (28) FIG. 6a shows the shaft of the bone punches displayed in FIGS. 1a to 1e and 3a to 3c, respectively, in the neutral position displayed in FIGS. 1b and 3b, respectively, in a distal detailed view.
- (29) FIG. 6b shows a detailed view of the first shaft part displayed in FIGS. 4a and 4b.
- (30) FIG. 6c shows the section of the second shaft part displayed in FIG. 5d with the locking area displayed in FIGS. 1a to 1c and 2a in an unlocking status in a top view.
- (31) FIG. 6d shows a section of the first shaft part displayed in FIG. 6b with a locking area in an unlocking status.
- (32) FIG. 6e shows an alternative to the first shaft part displayed in FIG. 6b.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(33) FIG. 1a shows a bone punches **100** with a proximally arranged locking area **230** in a punching position **206**. The bone punches **100** comprises a handle **300**. The handle **300** comprises a first handle portion **320** and a second handle portion **350**. The bone punches **100** further comprises a shaft **200**, which is shown to be in the punching position **206**. The shaft **200** comprises a first shaft portion **220** and a second shaft portion **250**. The locking area **230** is part of the second shaft part **250** and is in a neutral status **232**. The second shaft part **250** is located below the first shaft part **220** as illustrated. The second handle portion **350** is connected to the second shaft part **250**. The first handle portion **320** is movably, in this case rotatably, connected to the second shaft part **250** via a connecting device **310**. The bone punches **100** further comprises a return member **280** disposed in

the second shaft part **250** and oriented along a proximal-distal direction **400**. The return member **280** further comprises a spring **286**. In the area of the return member **280**, a partial section is shown in FIG. **1a**. The longitudinal direction of the shaft **200** is parallel to the proximal-distal direction **400**. The first shaft part **220** comprises a punching element **228** disposed at the distal end of the first shaft part **220**. The second shaft part **250** comprises a punch intake **258** disposed at the distal end of the second shaft part **250**. In the shown punching position **206**, the punching element **228** is positioned in or in contact with the punch intake **258**. The first shaft part **220** comprises two first guiding elements **221** arranged in the distal region immediately in front of the punching element **228**. The second shaft part **250** comprises two second guiding elements **251**, which are arranged in the distal area directly in front of the punch intake **258**.

(34) FIG. **1b** shows the bone punches **100** displayed in FIG. **1a** in a neutral position **204**. In the neutral position **204**, the first shaft part **220** is positioned further distally than in the punching position **206** (cf. FIG. **1a**). Accordingly, the punching element **228** is also spaced from the punch intake **258** so that tissue to be punched can be received in the distal shaft area. As in the punching position **206**, the locking area **230** is in the neutral status **232**. If the first handle portion **320** is deflected substantially proximally, i.e. along a first movement **322**, the shaft **200** is deflected into or towards the punching position **206**. The movement **322** follows a circular path, since the first handle portion **320**—as described above in connection with FIG. **1a**—is rotatably connected to the second shaft part **250** by means of the connecting device **310**. The rotation takes place around an axis that is located in the area of the connecting device **310**. If the first handle portion **320** is deflected distally, i.e. a second movement **324** is executed, the shaft **200** is moved into a disassembly position **202** (cf. FIG. **1c**). The second movement **324** is executed in the opposite direction to the first movement **322** and also follows a circular path or around the above-mentioned axis in the region of the connecting device **310**.

(35) FIG. **1c** shows the bone punches **100** displayed in FIGS. **1a** and **1b** in the disassembly position **202**. Along the proximal-distal direction **400**, the first handle portion **320** is maximally spaced from the second handle portion **350**. In the disassembly position **202**, the first shaft portion **220** is positioned away from the second shaft portion **250** along a vertical direction **404** that is perpendicular to the proximal-distal direction **400** and substantially along the longitudinal axis of the handle **300**. At the time of displacement of the shaft **200** from the neutral position **204** to the disassembly position **202**, the locking area **230** is deflected into an unlocking status **234** (cf. FIG. **6c**).

(36) FIG. **1d** shows the bone punches **100** displayed in FIGS. **1a** to **1c** in a top view. An intersection line is drawn centrally along the proximal-distal direction **400**.

(37) FIG. **1e** shows the bone punches **100** displayed in FIGS. **1a** to **1d** in the neutral position **204** displayed in FIG. **1b** in a sectional view **410-410** along the sectional line displayed in FIG. **1d**. The return member **280** (cf. FIG. **5e**) comprises a shaft **284** oriented along the proximal-distal direction **400**. The second shaft part **250** further comprises a guiding opening **274** in which the return member **280** is received or in which the shaft **284** is guided. The return member **280** further comprises a stop **282** which is oriented proximally. The return member **280** further comprises a spring **286** that is mounted on the shaft **284**. The first handle portion **320** comprises a handle head **340** in the upper region. The first shaft portion **220** comprises a guiding opening **229** in the proximal region (cf. FIGS. **2b**, **4b**). The first handle portion **320** further comprises a first load transfer area **344** arranged at the handle head **340**. The first load transfer area **344** is distally oriented and arranged in contact with the stop **282** such that by the first movement **322** (cf. FIG. **1b**), the return member **280** is moved distally into the guiding opening **274**. Furthermore, by the first movement **322** the upper end of the handle head **340** is moved distally, whereby the first shaft part **220** is also moved distally via the guiding opening **229** and finally the shaft **200** is moved into the punching position **202** (cf. FIG. **1a**).

(38) FIG. **2a** shows the first shaft part **220** of the bone punches **100** displayed in FIGS. **1a** to **1e** in a

side view. Opposite to the proximal-distal direction **400**, i.e. from distal to proximal, the first shaft part **220** further comprises, in addition to the two first guiding elements **221** in sequence, another four first guiding elements **222**, **223**, **224**, **225** and another two guiding elements **226**. The locking area **230** is located in the neutral status **232** and comprises a spring area **240** which comprises a fixed, i.e. non-deflectable, end towards the distal end and a loose, i.e. maximally deflectable, end towards the proximal end. The spring area **240** comprises a first spring arm **241** and a second spring arm **242**, which are arranged opposite each other (cf. FIG. **2b**). The second spring arm **242** is shown covered by the first spring arm **241** in FIG. **2a**. The spring area **240** comprises a recess **248** arranged at the fixed end of the spring area **240**. The spring area **240** is rounded or curved. The first spring arm **241** and the second spring arm **242** each comprise a projection **244**, wherein the projection **244** of the second spring arm **242** is covered herein by the projection **244** of the first spring arm **241**.

(39) FIG. **2b** shows a bottom view of the first shaft part **220** displayed in FIG. **2a**. The guiding opening **229** is continuous from bottom to top. The first spring arm **241** and the second spring arm **242** are each formed such that their loose ends, i.e., their proximal ends are deflectable along a horizontal direction **402**. The locking area **230** is located in the neutral status **232**.

(40) FIG. **2c** shows the second shaft part **250** of the bone punches **100** displayed in FIGS. **1a** to **1e** in a side view. Opposite the proximal-distal direction **400**, i.e. from distal to proximal, the second shaft part **250** comprises, in addition to the two second guiding elements **251** following one another, a further second guiding element **252**, a further second guiding element **253** and two further guiding elements **254**. The first guiding elements **221**, **222**, **223**, **224**, **225**, **226** and the second guiding elements **251**, **252**, **253**, **254** are configured to complement each other in such a way that the first shaft part **220** can be moved along the second shaft part **250** in the proximal-distal direction **400**, i.e. can be moved from the neutral position **204** to the punching position **206** and vice versa. The two first guiding elements **221** and the two second guiding elements **251** are engaged with each other in the punching position **202** and in the neutral position **204**, as are the first guiding element **222** and the second guiding element **252**, the first guiding elements **223**, **224**, **225** and the second guiding element **253**, and the two first guiding elements **226** and the two second guiding elements **254**. The second shaft part **250** further comprises a fixing area **260** that is complementary to the spring area **240** (cf. FIGS. **2a**, **2b**). The second shaft part **250** further comprises an intake **276**, which is configured complementary to the intake of the connecting device **310** (cf. FIG. **1a**).

(41) FIG. **2d** shows a top view of the second shaft part **250** displayed in FIG. **2c**. The fixing area **260** comprises two opposite bulges **262**.

(42) FIG. **2e** shows the first handle portion **320** of the bone punches **100** displayed in FIGS. **1a** to **1e** according to the orientation present at the disassembly position **202**. The first handle portion **320** comprises a first handle side **330** oriented distally and a second handle side **332** oriented proximally. On the first handle side **330**, the first handle portion **320** comprises a protrusion **326** that provides a more secure and ergonomic grip for the user during primary use of the bone punches **100**. Positioning of the user's index finger is intended above the protrusion **326**, and positioning of the user's middle, ring, and pinky fingers is intended below the protrusion.

(43) FIG. **2f** shows the handle head **340** of the first handle portion **320** displayed in FIG. **2e** in a side view. The handle head **340** comprises an intake **341**, which is configured for intake of the connecting device **310** (cf. FIG. **1a**) and is aligned with the intake **276** of the second shaft part **250** (cf. FIG. **2c**). In the upper region, the handle head **340** further comprises a second load transfer area **346** and a third load transfer area **348**. The second load transfer area **346** is oriented distally and is configured such that during the first movement **322** (cf. FIG. **1b**) load can be transferred to the distal wall of the guiding opening **229** (cf. FIG. **2b**), which causes the first shaft part **220** to move distally. The third load transfer area **348** is oriented proximally and is configured to transfer load to the proximal wall of the guiding opening **229** (cf. FIG. **2b**), causing the first shaft part **220** to move

proximally and to move from the neutral position **204** to the disassembly position **202**. The head area **340** further comprises a recess **342** that reduces moving masses, which can improve the handling properties of the bone punches **100**. The first load transfer area **344**, the second load transfer area **346** and the third load transfer area **348** are each bent or curved. This ensures that no tilting occurs when transferring the load from the first handle portion **320** along a circular path (cf. FIG. **1b**) to the first shaft part **220** and to the return member **280**, where an essentially linear movement is generated, and that the handle head **340** rolls on the first shaft part **220** and on the return member **280**. This may increase the robustness of the bone punches **100** and improve its handling properties.

(44) FIG. **2g** shows the first handle portion **320** displayed in FIG. **2e** in a front view, i.e. against the proximal-distal direction **400**. The first handle side **330** is broader than the handle head **340**. The handle head **340** is narrow enough to be accommodated in the second shaft part **250** (cf. FIGS. **2c**, **2d**).

(45) FIG. **2h** shows the second handle portion **350** of the bone punches **100** displayed in FIGS. **1a** to **1e** in a side view. The second handle portion **350** comprises a first handle side **360** oriented proximally and a second handle side **362** oriented distally. On the first handle side **360**, the second handle portion **350** comprises a protrusion **356** that provides a more secure and ergonomic grip for the user during the main use of the bone punches **100**. The protrusion **356** is configured to prevent the user's hand from sliding upward toward the proximal end of the shaft **200**. For this purpose, the user's metacarpal region is intended to rest between the thumb and index finger immediately below the protrusion **356**. The second handle portion **350** further comprises a plug element **352**, which is arranged in the upper region of the second handle portion **350** and by means of which the second handle portion **350** is connected or joined to the second shaft part **250** (cf. FIG. **1a**).

(46) FIG. **2i** shows the second handle portion **350** displayed in FIG. **2h** in an oblique view from above. The first handle side **360** is oriented upward in the illustration and the second handle side **362** is oriented downward. The plug element **352** comprises an oval cross-section. Thus, a good assemblability of the second handle portion **350** and the second shaft part **250** can be achieved. It is also possible that the plug element **352** comprises other cross-sections that, for example, support or improve the load transfer between the second shaft part **250** and the second handle portion or reduce deformation between these two components. Such cross sections may be, e.g., round, polygonal, but also cross- or star-shaped cross sections.

(47) FIG. **3a** shows another bone punches **100** with a distally arranged locking area **230** in a punching position **202**. In the area of the locking area **230**, the bone punches **100** is shown in partial section. The first spring arm **241** and the second spring arm **242** are arranged one behind the other along the proximal-distal direction **400**, with the first spring arm **241** being arranged more distally than the second spring arm **242**. The locking area **230** and the fixing area **260** are arranged to engage with each other. The fixing area **260** comprises two cylinders **268**. The first spring arm **241** and the second spring arm **242** each comprise a bulge **246**. The two cylinders **268** are located at the proximal end of each of the first and second spring arms **241**, **242** and at their fixed ends, respectively.

(48) FIG. **3b** shows the bone punches **100** displayed in FIG. **3a** in a neutral position **204**. In the area of the locking area **230**, the bone punches **100** is shown in partial section. The two cylinders **268** are each located immediately in front of the two bulges **246**, i.e. further proximally than the two bulges **246**. Accordingly, before the first shaft part **220** is displaced further proximally, the two cylinders **268** would have to overcome the two bulges **246**. This would require the first and second spring arms **241**, **242** to be deflected along the vertical direction **404**, again creating space for the two cylinders **268** to slide past the two bulges **246**. This situation is shown in FIG. **3c**.

(49) FIG. **3c** shows the bone punches displayed in FIGS. **3a** and **3b** in a disassembly position **202**. In the area of the locking area **230**, the bone punches **100** is shown in partial section. The two cylinders **268** are each located distally from the respective bulges **246** associated with them. The

first shaft part **220** is located vertically away from the second shaft part **250**, which is achieved by the first and second spring arms **241**, **242** each being configured to taper distally along the vertical direction **404** and the respective portion of the first shaft part **220** located correspondingly above the first and second spring arms **241**, **242** being configured to thicken to the same extent.

(50) FIG. **4a** shows a first shaft part **220** of the bone punches **100** displayed in FIGS. **3a** to **3c** in a side view. The first shaft part **220** shown in FIG. **4a** is configured similarly to the first shaft part **220** displayed in FIGS. **2a** and **2b**, with the first shaft part **220** displayed in FIG. **4a** comprising only the first guiding elements **221**, **222**, **223**, **226**. Instead of the first guiding elements **224**, **225** in the first shaft part **220** displayed in FIGS. **2a** and **2b**, the first and second spring arms **241**, **242** and the locking area **230** are provided in the first shaft part **220** displayed in FIG. **4a**.

(51) FIG. **4b** shows the first shaft part **220** displayed in FIG. **4a** from below. The first and second spring arms **241**, **242** are centred in the first shaft part **220** with respect to the horizontal direction **402**. The first and second spring arms **241**, **242** are furthermore configured in such a way that they—like the first guiding elements **224**, **225** of the first shaft part **220** displayed in FIGS. **2a** and **2b**—promote guidance of the first shaft part **220** along the proximal-distal direction **400**.

(52) FIG. **4c** shows a second shaft part **250** of the bone punches **100** displayed in FIGS. **3a** to **3c** in a side view. Compared to the second shaft part **250** displayed in FIGS. **2c** and **2d**, the second shaft part **250** displayed in FIG. **4c** additionally comprises the two cylinders **268** arranged in the second guiding element **253**.

(53) FIG. **4d** shows the second shaft part **250** displayed in FIG. **4c** in a top view. The two cylinders **268** are aligned such that their longitudinal axes run along the horizontal direction **402**, i.e. vertical to the proximal-distal direction **400**.

(54) FIG. **5a** shows an exploded view of a connecting device **310** of the bone punches **100** displayed in FIGS. **1a** to **1e** and **3a** to **3b**, respectively. The connecting device **310** comprises a connecting insert **312** and a counter part **316**. The connecting insert **312** comprises a stop **314**. The counter part **316** comprises a stop **318**, too. The counter part **316** further comprises an inlet **319** into which the connecting insert **312** is insertable. The connecting insert **312** can be inserted until its stop **314** abuts the end of the intake **319** opposite the stop **318** of the counter part **316**. The connecting device **310** can in particular be provided with a screw device, i.e. the connecting insert **312** further comprises an external thread and the intake **319** a matching internal thread. In this way, a variable and easy-to-use connecting device can be provided. However, a press connection can also be provided for the connecting device **310**, i.e. the intake **319** comprises a smaller inner diameter than the connecting insert **312**. Furthermore, a welded or riveted connection can also be provided for the connecting device **310**.

(55) FIG. **5b** shows the connecting device **310** displayed in FIG. **5a** in the assembled state. The connecting device **310** is configured so that the stops **314**, **318** are in clearance-free or nearly clearance-free contact with the outside of the second shaft part **250** in the assembled state. The Intake **319** is configured to support the second shaft part **250** and the first handle portion **320**. For this purpose, the outer contour of the intake **319** is configured complementary to the inner contour of the intake **276** of the second shaft part **250** (cf. FIG. **2c**) and to the inner contour of the intake **341** of the first handle portion **320** (cf. FIG. **2f**) in such a way that, on the one hand, the first handle portion **320** is securely mounted in the second shaft part **250** and, on the other hand, the first handle portion **320** can be rotated with low resistance and low clearance or without clearance in the second shaft part **250**.

(56) FIG. **5c** shows a section of the second shaft part **250** with a first stop element **270** in a top view. Although the figure shows a detail of the second shaft part **250** displayed in FIGS. **2c** and **2d**, it can also be applied to the second shaft part **250** displayed in FIGS. **4c** and **4d**. The first stop element **270** is configured for this purpose and arranged in such a way that the movability of the first handle portion **320**, in particular the handle head **340**, is restricted proximally in the course of the second movement **324**. This prevents damage or jamming of the components that move against

each other, in particular the first handle portion **320** and the first shaft part **220**.

(57) FIG. **5d** shows a section of the second shaft part **250** with a second stop element **272** in a top view. Although the figure shows a detail of the second shaft part **250** displayed in FIGS. **2c** and **2d**, it is equally transferable to the second shaft part **250** displayed in FIGS. **4c** and **4d**. The second stop element **272** is configured for this purpose and arranged in such a way that the movability of the first handle portion **320**, in particular of the handle head **340**, is restricted distally in the course of the first movement **322**. This prevents damage or jamming of the components that move against each other, in particular the first handle portion **320** and the first shaft part **220**. FIG. **5d** also shows details of the fixing area **260** and the bulges **262**. The two bulges **262** each comprise a proximal edge **264** and a distal edge **266**. The two distal edges **266** are formed in such a way that in the neutral position **204** (cf. FIG. **1b**), when the projections **244** are guided from distal to proximal in the fixing area **260** in the course of the second movement **324**, the projections **244** rest against the distal edges **266** and a further movement of the first shaft part **220** towards the proximal is prevented for the moment. If sufficient force is then applied and the second movement **324** continues, the projections **244** travel along the distal edges **266**, causing the first and second spring arms **241**, **242** to deflect along and against the horizontal direction **402**, respectively. In this way, the unlocking status **234** can be achieved, i.e. the projections **244** can overcome the distal edges **266** and slide along the proximal edges **264** in a proximal direction, causing the shaft **200** to be moved into the disassembly position **202** (cf. FIG. **1c**). If the shaft **200** is then to be moved back to the neutral position **204** against the second movement **324**, sufficient force must be applied so that the projections **244** can overcome the proximal edges **264**. The two proximal edges **264** comprise a smaller slope along the proximal distal direction **400** than the two distal edges **266**. This makes it possible to provide greater resistance to moving the shaft **200** from the neutral position **204** to the disassembly position **202** than to moving the shaft **200** from the disassembly position **202** to the neutral position **204**. Such proximal and distal edges **264**, **266** can also be provided in the first shaft part **220** displayed in FIGS. **4a** and **4b** (cf. FIG. **6b**).

(58) FIG. **5e** shows a return member **280** in a side view. The stop **282** is further configured to limit distal penetration of the return member **280** into the guiding opening **274** (cf. FIG. **1e**).

(59) FIG. **5f** shows a side view of the spring **286** associated with the return member **280** displayed in FIG. **5e**. The spring **286** is placed on the shaft **284** (cf. FIG. **1e**) and is configured to provide resistance, in particular increasing resistance, to the first movement **322**. Furthermore, with the spring **286** it can be achieved that during the main use of the bone punches **100** no force is applied by the operator to reset or move back the shaft **200** from the punching position **206** to the neutral position **204**, which can increase the handling.

(60) FIG. **6a** shows the shaft **200** of the bone punches **100** displayed in FIGS. **1a** to **1e** and **3a** to **3c**, respectively, in the neutral position **204** displayed in FIGS. **1b** and **3b**, respectively, in a distal detailed view. By means of the cooperation of the two first guiding elements **221** arranged at the distal end with the two second guiding elements **251** arranged at the distal end, secure guiding of the first shaft part **220** in the outer region is made possible, and by means of the cooperation of the first guiding element **222** arranged a bit further proximally and second guiding element **252**, secure guiding in the central region is enabled (cf. FIGS. **2a** to **2d**). Thus, the punching element **228** and the punch intake **258** can be moved towards each other precisely and safely, especially under load.

(61) FIG. **6b** shows a detailed view of the first shaft part **220** displayed in FIGS. **4a** and **4b**. The two bulges **246** each comprise a proximal edge **264** and a distal edge **266**, wherein in the first shaft part **220** displayed in FIG. **6b**, the two proximal edges **264** are overcome with the cylinders **268** to move the shaft **200** into the disassembly position **202**, but not the two distal edges **266**, as in the second shaft part **250** displayed in FIGS. **5d** and **2c** and **2d**.

(62) FIG. **6c** shows the section of the second shaft part **250** displayed in FIG. **5d** with the locking area **230** displayed in FIGS. **1a** to **1c** and **2a** in the unlocking status **234** in a top view. The first spring arm **241** and the second spring arm **242** are oriented in such a way that the two projections

244 are each located on the tip of the two bulges **262**, i.e., between the proximal edge **264** and the distal edge **266**. As a result, the first spring arm **241** is oriented opposite to the horizontal direction **402**, i.e., inward, and the second spring arm **242** is deflected with the horizontal direction **402**, i.e., also inward, allowing the first shaft part **220** to be deflected or displaced further proximally, i.e., against the proximal-distal direction **400**. This allows the shaft **200** to be moved from the neutral position **204** to the disassembly position **202** (cf. FIG. 1c).

(63) FIG. 6d shows a section of the first shaft part **220** displayed in FIG. 6b with the locking area **230** in the unlocking status **234**. The second spring arm **242** is shown, wherein the illustration and what is described in this context can be applied in the same way to the first spring arm **241**. The cylinder **268** is located on the tip of the bulge **246**, i.e., between the proximal edge **264** and the distal edge **266**, causing the second spring arm **242** to deflect along the vertical direction **404**, thereby moving the shaft **200** from the neutral position **204** to the disassembly position **202** (cf. FIGS. 3b and 3c).

(64) FIG. 6e shows an alternative to the first shaft part **220** displayed in FIG. 6b. The two bulges **246** shown here are more voluminous and each comprise a proximal edge **264** and a distal edge **266** with a smaller pitch than the two bulges **246** or proximal edges **264** and distal edges **266**, respectively displayed in FIG. 6b. The lower slope of the proximal edges **264** can provide less resistance to moving the locking area **230** to the unlocking position **234**, thereby facilitating handling. With the increased volume of the two bulges **246**, it can be achieved that their lifetime is increased.

REFERENCE SIGNS LIST

(65) **100** bone punches **200** shaft **202** disassembly position **204** neutral position **206** punching position **220** first shaft part **221** first guiding element **222** first guiding element **223** first guiding element **224** first guiding element **225** first guiding element **226** first guiding element **228** punching element **229** guiding opening **230** locking area **232** neutral status **234** unlocking status **240** spring area **241** first spring arm **242** second spring arm **244** projection **246** bulge **248** undercut **250** second shaft part **251** second guiding element **252** second guiding element **253** second guiding element **254** second guiding element **258** punch intake **260** fixing area **262** bulge **264** proximal edge **266** distal edge **268** cylinder **270** first stop element **272** second stop element **274** guiding opening **276** intake **280** return member **282** stop **284** shaft **286** spring **300** handle **310** connecting device **312** connecting insert **314** stop **316** counter part **318** stop **319** intake **320** first handle portion **322** first movement **324** second movement **326** protrusion **330** first handle side **332** second handle side **340** handle head **341** intake **342** recess **344** first load transfer area **346** second load transfer area **348** third load transfer area **350** second handle portion **352** plug element **356** protrusion **360** first handle side **362** second handle side **400** proximal-distal direction **402** horizontal direction **404** vertical direction **410-410** sectional view

Claims

1. A disassemblable bone punch comprising a shaft and a handle, wherein the shaft comprises a first shaft part and a second shaft part, wherein the first shaft part comprises at least one first guiding element, and the second shaft part comprises at least one second guiding element, wherein the first shaft part is connected to the second shaft part such that the first guiding element is in engagement with the second guiding element and the first shaft part is movable relative to the second shaft part parallel to a proximal-to-distal direction relative to the second shaft part, wherein the shaft is movable into a disassembly position, into a neutral position and into a punching position, wherein the handle comprises a first handle portion and a second handle portion, wherein the first handle portion is movably connected to the second shaft part, wherein the second handle portion is connected to the second shaft part, wherein by a first movement of the first handle portion the shaft is displaceable from the neutral position into the punching position, wherein by a

second movement of the first handle portion the shaft is displaceable from the neutral position into the disassembly position, wherein the first shaft part comprises a locking area, and wherein the locking area is movable from a neutral status to an unlocking status, wherein the locking area is configured such that, when the locking area is in the neutral status, moving of the shaft into the disassembly position is prevented and, by the second movement, the locking area is simultaneously moved into the unlocking status and the shaft is moved into the disassembly position; and wherein the locking area comprises a deflectable spring area, and wherein the locking area is movable into the unlocking status by deflecting the spring area.

2. The disassemblable bone punch according to claim 1, wherein the spring area is configured to be deflectable substantially perpendicular to the proximal-to-distal direction.

3. The disassemblable bone punch according to claim 2, wherein the spring area is deflectable substantially parallel to a horizontal direction.

4. The disassemblable bone punch according to claim 2, wherein the spring area is deflectable substantially parallel to a vertical direction.

5. The disassemblable bone punch according to claim 1, wherein the spring area comprises a recess.

6. The disassemblable bone punch according to claim 1, wherein the second shaft part comprises a fixing area, the fixing area being configured and arranged to interact with the locking area such that if the locking area is in the neutral status, additional resistance is provided by the fixing area opposing the second movement of the shaft into the disassembly position.

7. The disassemblable bone punch according to claim 6, wherein the locking area comprises at least one first resistance element and/or the fixing area comprises at least one second resistance element, wherein the first resistance element and/or the second resistance element is/are arranged and configured such that an additional resistance opposes the movement of the locking area into the unlocking status.

8. The disassemblable bone punch according to claim 7, comprising a first resistance element and a second resistance element, wherein the first resistance element and the second resistance element are interactively arranged and configured such that additional resistance opposes the movement of the locking area into the unlocking status.

9. The disassemblable bone punch according to claim 8, wherein the first resistance element is configured as a projection and/or the second resistance element is configured as a bulge.

10. The disassemblable bone punch according to claim 8, wherein the first resistance element is configured as a bulge and/or the second resistance element is configured as a cylinder.

11. The disassemblable bone punch according to claim 1, wherein the bone punch further comprises a first stop element preferably arranged in the second shaft part, wherein the first stop element is configured and further arranged such that a movability of the first handle portion is limited proximally.

12. The disassemblable bone punch according to claim 1, wherein the bone punch further comprises a second stop element arranged in the second shaft portion, wherein the second stop element is configured and further arranged such that a movability of the first handle portion is limited distally.

13. The disassemblable bone punch according to claim 1, wherein the bone punch further comprises a return member preferably arranged in the second shaft part, wherein the return member is configured and further arranged such that after performing the first movement, the shaft is automatically displaced from the punching position back to the neutral position and, the first movement is opposed by an increasing resistance.
